

Revised Generation Interconnection System Impact Study Report

For Queue Project Q12-0178

Grays River 138 kV
January 28, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0178. The Interconnection Customer proposes a 164.8 MW natural gas combined cycle facility interconnecting at the Grays River 138 kV substation in the Pennsylvania area. The request was received on November 25, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 136 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$10,338,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of CORONET-FT HUMBUG 500kV Line, the MTN 500-345 kV XFMR loads to 118.6% of its 876 MVA rating. Project DFAX: 5.1%.

For the loss of WESTWOOD GENERATION LLC-CORONET 500kV, the TALENENERGY MARTINS CRK LLC HARRISBURG 500 kV/345 kV Autotransformer loads to 104.8% of its 982 MVA rating. Project DFAX: 7.6%.

For the loss of PINEY CREEK PROJECT-HILLTOP 500kV, the PINEY CREEK PROJECT-CALDWELL 345kV loads to 104.6% of its 1,118 MVA rating. Project DFAX: 9.5%.

The AMERICAN-URBAN 345kV is overloaded to 103.1% of its 1,605 MVA emergency rating for the loss of the ORCHARD RD-FORT HUMBUG 500kV. The project contributes 5.1% DFAX (8.5 MW) to this overload.

For the loss of FORT HUMBUG-ORCHARD ROAD 500kV, the ORCHARD RD 500 kV/345 kV Autotransformer loads to 100.6% of its 684 MVA rating. Project DFAX: 5.5%.

The CARBON CENTER 500/345kV TX is overloaded to 100.4% of its 851 MVA emergency rating for the loss of the BLACKROCK SUBSTATION-SHAWVILLE STA SUBSTATION 345kV Line. The project contributes 10.7% DFAX (17.7 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
MOUNTAIN 500-345 kV Transformer	Replace transformer with higher-rated unit	\$23,558,000	10% (\$2,236,000)
TALENENERGY MARTINS CRK LLC HARRISBURG 500/345kV XFMR	Replace transformer with higher-rated unit	\$21,189,000	4% (\$931,000)
CALDWELL-PINEY CRK PROJECT 345kV	Reconductor line	\$36,295,000	5% (\$1,908,000)
URBAN-AMERICAN 345kV	Reconductor line	\$21,502,000	5% (\$1,142,000)
ORCHARD ROAD 500 kV/345 kV Transformer Bank	Replace transformer with higher-rated unit	\$36,346,000	5% (\$1,700,000)
CARBON CENTER 500/345kV TX	Replace transformer with higher-rated unit	\$38,357,000	6% (\$2,421,000)
	Total allocated to project		\$10,338,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.